Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/02/2006
PI Name:	Swager, Timothy Ph.D.		
Project Title:	Photonic Technologies for Noninvasive Detection, Diagnosis, and Treatment of Cancer		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Biomolecular physics & chemistry		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	02139	**Congressional District:**	8
Comments:			
Project Type:	INTERDISCIPLINARY (2004-2007)	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	12/01/2001	**End Date:**	11/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAS2-02056		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Self-amplifying (SA) semiconductive organic sensory polymers and associated photonic technologies are being developed for the detection and treatment of cancer. The ability of these polymers to transport energy (excitons) over large distances has been demonstrated by the MIT team to produce dramatic amplification of sensory responses. A microsphere platform technology will be developed that utilizes and enhances the SA nature of these materials, their low lasing thresholds, and their large two-photon absorption cross-sections. A variety of transduction methods are under investigation as are methods for the up-conversion of light for photodynamic treatment. The MIT team is focused on the development of novel optical materials and devices for the detection, diagnosis, and treatment of cancer. To obtain performance beyond conventional solutions we are developing devices based upon new generations of semiconducting polymers that have non-linear absorption and emission properties. These features lead to ultrasensitive sensors and the ability to create materials for the up-conversion of photons.		

	This team's approach is not simply an incremental advancement of fiber-optic sensor and photodynamic therapy technologies. The sensors are based upon semiconductive organic polymers that have sensitivities multiple orders of magnitude beyond those provided by conventional methods. The enhancements will be provided by integrating concepts first reported by Swager in 1995, wherein he demonstrated that semiconducting polymers are capable of a signal-amplifying (SA) sensory response. Swager's SA sensory technology has broad implications, and groups throughout the world have confirmed the utility of this method.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This contract has provided a foundation for a long and fruitful application of amplifying fluorescent polymers to biomedical problems.
Task Progress:	This contract has resulted in key advanced in technologies for the Noninvasive Detection, Diagnosis, and Treatment of Cancer that apply the amplifying properties of electronic polymers. The best measure of the success is that 4 patents have been filed and licensed. There are ongoing efforts to further mature these technologies by the industrial licensee, Nomadics Inc. The specific technological successes are: 1. Laser Sensors for Ultra Trace Detection of Nitroaromatics 2. Polymers for Direct Protein Detection 3. Direct Detection of Live Cells by Fluorescent Polymers 4. Far Red Dyes for in vivo Diagnostics There are multiple inventions that are in their final stages of refinement. Protease detection has been demonstrated in homogenous solution and the amplification from the electronic polymers have been quantified. Based upon a considerable amount of basic research over the last 3 years we have converged on the optimal design for protease detection based upon carefully structured microspheres. For the detection of biological molecules, organisms, and activity we have strived to produce new schemes for the energy transfer to longer wavelength fluorescent acceptor dyes. This scheme, which before this contract appeared straightforward, was very complex and has resulted in our realization that for conjugated polymers with delocalized and mobile excited states the conventional rules that govern fluorescent resonant energy transfer (FRET) do not apply. We demonstrated this decisively and have been working to exploit a new paradigm for bio-sensing that is much more powerful than conventional FRET. Most importantly our new model indicates that there need not be any spectral overlap between the donor and acceptor chromophores. To best realize this principle we have embarked on the synthesis of far red dyes, that are novel and useful in their own right, as acceptors. We expect in the very near term to use these dyes to create new emissions that are far to the red of the residual polymer emission. This results in a dark background at the detection wavelength and provides massively better signal to noise for superior detection limits. We have also found important new associations between polymers that can result in dramatic red shifted emissions. These shifts are intrinsically amplified by the energy migration in the polymers and we are finishing the development of liposome platforms for fielding of this technology. We expect these additional inventions will be ready for transition to industry in the spring. This contract has provided a foundation for a long and fruitful application of amplifying fluorescent polymers to biomedical problems. Key technology has been developed and is undergoing commercialization. Given that it has only been 3 years in the making, this alone is a clear success. The PI is actively pursuing additional applications and follow-on funding from other government agencies and private sources to further develop fundamental understanding and science that will impact health care.
Bibliography Type:	Description: (Last Updated: 09/23/2020)
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